

Technical Design Document (TDD)

Project: HireSense (AI-Powered Applicant Tracking System)

Document Version: 1.0.0

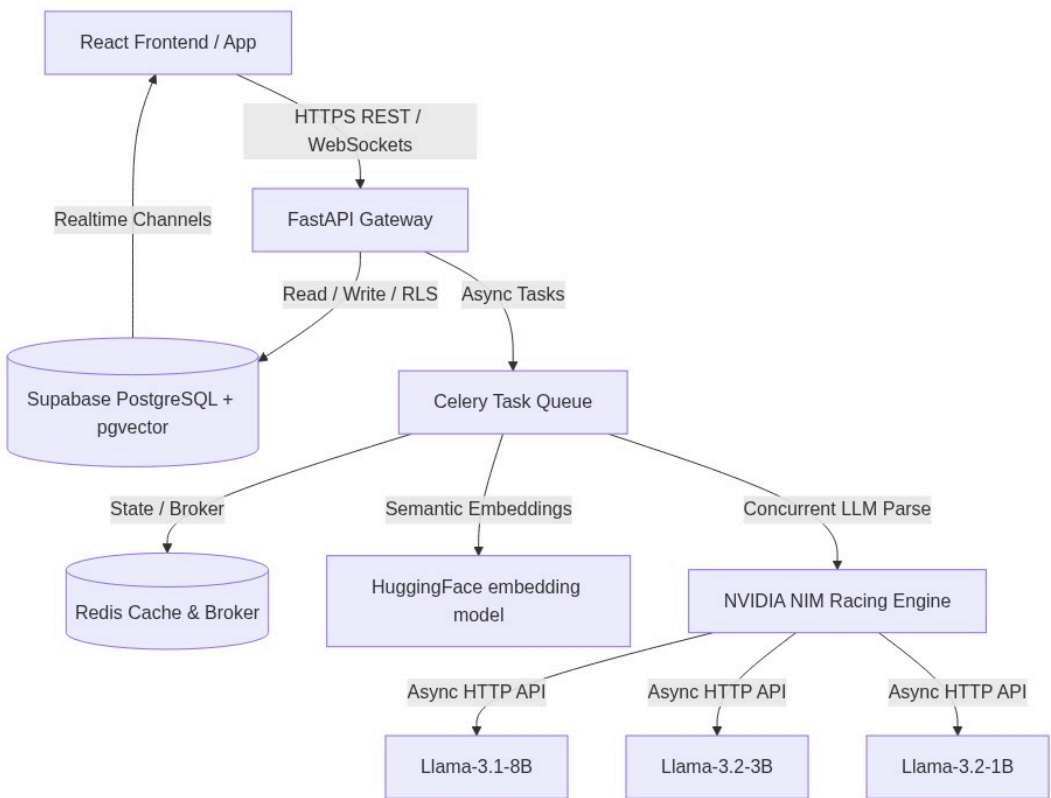
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Status: Approved

Target Architecture: Supabase + pgvector + FastAPI + NVIDIA NIM Racing Engine

1. System Overview & High-Level Architecture

HireSense is designed to handle high-throughput resume processing, semantic candidate matching, and real-time application tracking. The platform follows a decoupled, service-oriented architecture comprising a responsive frontend dashboard, a high-performance asynchronous API backend, a Postgres vector database, and a highly concurrent AI inference pipeline.



1.1 Key Architectural Components

1. Frontend Application: Built using React 19 and Vite, styled with vanilla CSS custom properties for absolute design control and a high-fidelity cinematic look. Connects to FastAPI for business logic and directly to Supabase WebSockets for real-time dashboard sync.
2. FastAPI Application Gateway: Serving REST endpoints and orchestrating backend services. Utilizes Uvicorn

as an ASGI server for high-concurrency connection pooling.

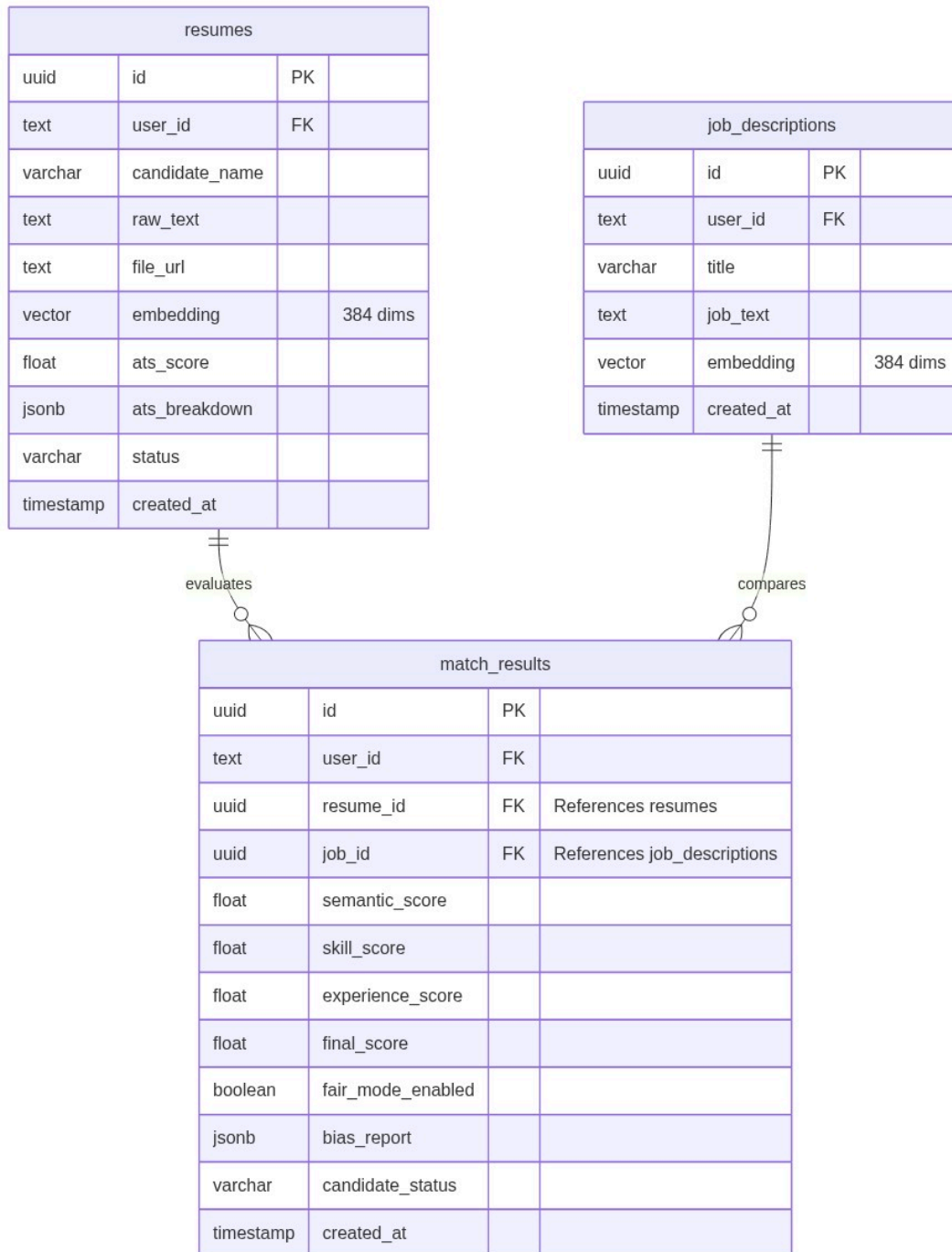
3. Supabase PostgreSQL Database: Empowered by pgvector for storing dense sentence embeddings (384 dimensions) and utilizing standard relational tables.

4. Celery Asynchronous Workers: Using Redis as a message broker to execute long-running resume parsing, embedding generation, and LLM scoring completely out-of-band.

5. NVIDIA NIM Racing Engine: A custom low-latency API router that queries multiple model sizes concurrently, parsing the fastest valid output to minimize user wait times.

2. Database Schema & Vector Search Design

The database layer utilizes Supabase (PostgreSQL 15+) with pgvector enabled. Strict row-level security (RLS) is applied to all tables to ensure data isolation.



2.1 pgvector Configuration & HNSW Indexing

To support sub-second query execution across thousands of candidate profiles, dense embeddings are computed using the all-MiniLM-L6-v2 model (384 dimensions). A Hierarchical Navigable Small World (HNSW) index is created on the vector columns using cosine_distance (<=> operator):

```
-- Enable the pgvector extension
CREATE EXTENSION IF NOT EXISTS vector;

-- Create high-performance HNSW index for resumes
CREATE INDEX ON resumes USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
```

```
-- Create high-performance HNSW index for job descriptions
CREATE INDEX ON job_descriptions USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);
```

2.2 Dynamic Row-Level Security (RLS) Policies

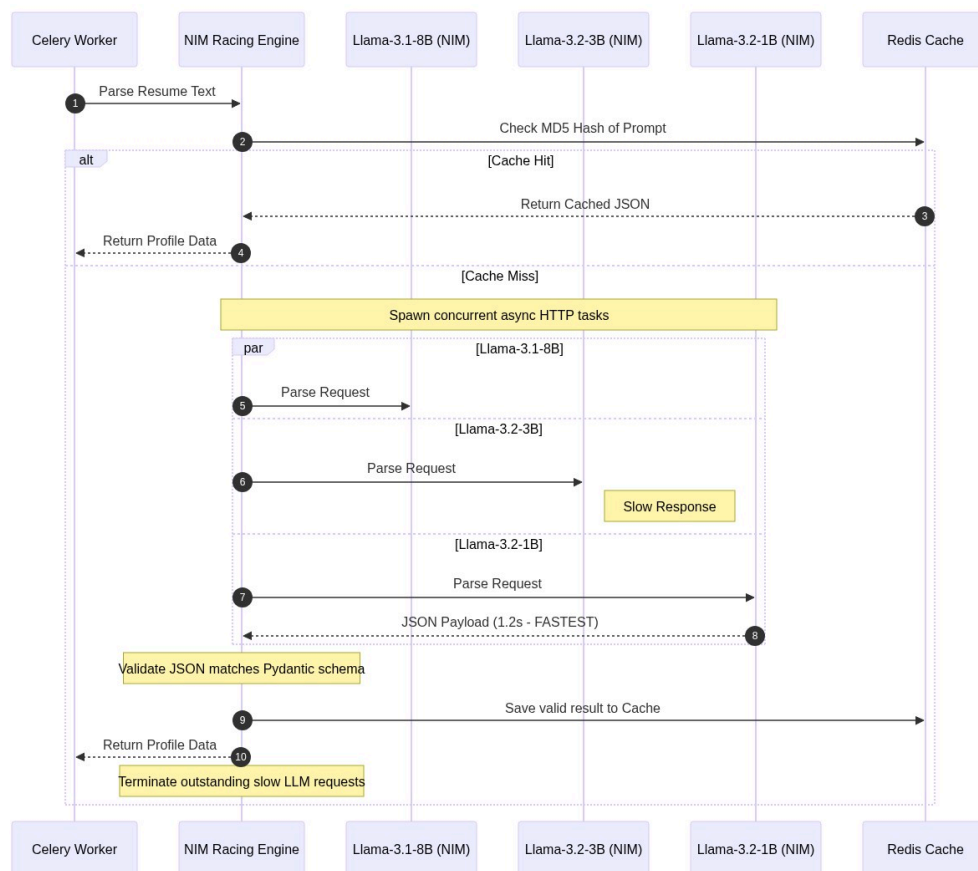
Data privacy is enforced directly within the database layer. Every table is bound to the authenticated `user_id` of the recruiter or applicant:

```
ALTER TABLE resumes ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Users can only read and write their own resumes"
ON resumes FOR ALL
TO authenticated
USING (auth.uid()::text = user_id)
WITH CHECK (auth.uid()::text = user_id);
```

3. Multi-Model NVIDIA NIM Racing Engine

The parser uses a multi-model asynchronous competitive API runner (NIM Racing Engine). When a resume needs parsing, the application executes requests to three NVIDIA NIM endpoints concurrently. The fastest model returning a JSON matching our Pydantic schema wins, while slow or failed requests are discarded.



3.1 Asynchronous Execution Code Design

Using python's asyncio and httpx to trigger simultaneous calls:

```
import asyncio
import httpx
import hashlib
import json

class NIMRacingEngine:
    def __init__(self, endpoints: dict, api_key: str):
        self.endpoints = endpoints
        self.api_key = api_key
        self.headers = {
            "Authorization": f"Bearer {api_key}",
            "Content-Type": "application/json"
        }

    async def query_model(self, client: httpx.AsyncClient, name: str, url: str, payload: dict) -> dict:
        try:
            response = await client.post(url, json=payload, headers=self.headers, timeout=8.0)
            if response.status_code == 200:
                data = response.json()
                # Local validation hook
                if "candidate_name" in data and "skills" in data:
                    return {"model": name, "data": data, "success": True}
                raise ValueError(f"Invalid output from {name}")
            except Exception as e:
                return {"model": name, "error": str(e), "success": False}

    async def race_inference(self, prompt: str) -> dict:
        # Prompt hashing
        prompt_hash = hashlib.md5(prompt.encode('utf-8')).hexdigest()

        payload = {
            "model": "meta/llama3",
            "messages": [{"role": "user", "content": prompt}],
            "temperature": 0.1,
            "response_format": {"type": "json_object"}
        }

        async with httpx.AsyncClient() as client:
            tasks = [
                self.query_model(client, name, url, payload)
                for name, url in self.endpoints.items()
            ]

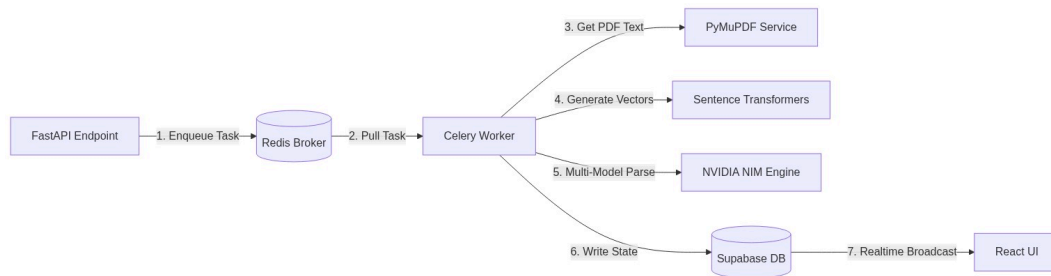
            # Use asyncio.as_completed to get the fastest resolver first
            for completed_task in asyncio.as_completed(tasks):
                result = await completed_task
                if result["success"]:
                    # Immediately return the fastest successful result
                    return result

            # Local fallback parser in case all API calls fail
            return self.local_heuristic_fallback(prompt)
```

```
def local_heuristic_fallback(self, prompt: str) -> dict:
    # Minimalist regular-expression parsing logic
    return {
        "model": "local_fallback",
        "data": {
            "candidate_name": "Unknown Candidate",
            "skills": ["Python", "FastAPI"],
            "experience": 0
        }
    }
```

4. Asynchronous Pipeline & Redis Caching

To ensure the frontend remains highly responsive, heavy processing is offloaded to Celery workers using Redis as the message broker.



4.1 Resume Upload to Processing Sequence

1. Upload: React frontend posts the resume PDF file to `/api/v1/resumes/upload`.
2. Ingest: FastAPI writes the file metadata to the database with state Pending, copies the binary to Supabase Storage, and schedules a Celery task:

```
process_resume.delay(resume_id=new_resume_id, file_url=storage_url).
```

3. Immediate Acknowledge: FastAPI returns an HTTP 202 accepted response in `< 150ms`.

4. Execution:

- The worker parses the PDF raw text using PyMuPDF.
- Text embeddings are computed in an isolated HuggingFace thread.
- The NIM Racing Engine processes the text for entities.

5. Real-time Notify: The worker updates the database record. A database trigger publishes the state change (Screened) which streams over WebSockets directly to the recruiter's Kanban board.

5. Compliance & Evaluation Algorithm

The evaluation engine leverages both mathematical semantic similarities and traditional heuristic taxonomies.

5.1 The Weighting Formulas

When comparison is initiated between a candidate's resume and a job description:

$$\text{Final Score} = w_1 \cdot S_{\text{semantic}} + w_2 \cdot S_{\text{skill}} + w_3 \cdot S_{\text{experience}} + w_4 \cdot S_{\text{formatting}}$$

Standard Operational Mode Weights

- Semantic Vector Match ($w_1 = 0.40$): The cosine similarity between the Job Description embedding V_{jd} and Resume embedding V_{resume} :

$$S_{\text{semantic}} = \frac{V_{\text{jd}} \cdot V_{\text{resume}}}{\|V_{\text{jd}}\| \|V_{\text{resume}}\|}$$

- Skill Overlap Match ($w_2 = 0.30$): Standard taxonomy matching:

$$S_{\text{skill}} = \frac{|\text{Skills}_{\text{resume}} \cap \text{Skills}_{\text{jd}}|}{|\text{Skills}_{\text{jd}}|}$$

- Experience Gap Fit ($w_3 = 0.20$): Heuristic comparison:

$$S_{\text{experience}} = 1.0 - \min\left(1.0, \frac{|\text{Exp}_{\text{required}} - \text{Exp}_{\text{candidate}}|}{|\text{Exp}_{\text{required}}|}\right)$$

- Resume Strength Heuristic ($w_4 = 0.10$): Layout completeness score based on structure, section headers, and text formatting.

NYC LL144 "Fair Mode" Weights (Bias Reduction Mode)

When Fair Mode is toggled, structural experience age checks are completely bypassed to eliminate chronological grading bias for students:

- Semantic Vector Match: $w_1 = 0.55$
- Skill Overlap Match: $w_2 = 0.35$
- Resume Strength Heuristic: $w_4 = 0.10$
- Experience Gap Fit: $w_3 = 0.00$

6. Security, Auditing & Disaster Recovery

6.1 Path Traversal Hardening

Files processed by PyMuPDF are heavily sanitized. The backend uses Python's `werkzeug.utils.secure_filename` to ensure filenames are strictly restricted to alphanumeric strings, preventing arbitrary file write injection:

```
import os
from werkzeug.utils import secure_filename

def save_and_sanitize_upload(upload_file, upload_dir) -> str:
    cleaned_name = secure_filename(upload_file.filename)
    # Block path traversal attempts (e.g. ../../etc/passwd)
    full_path = os.path.abspath(os.path.join(upload_dir, cleaned_name))
    if not full_path.startswith(os.path.abspath(upload_dir)):
        raise ValueError("Security violation: Path traversal detected!")
    return full_path
```

6.2 Data Redundancy & Daily Cloudflare R2 Backups

The database is backed up daily via an automated worker pipeline. A shell script executes a schema and data dump, encrypts the compressed archive via AES-256, and pushes the payload directly to a secure, private bucket on Cloudflare R2:

```
#!/bin/bash
# Backup script executed daily via cron
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
BACKUP_NAME="hiresense_backup_${TIMESTAMP}.sql.gz"
ENC_BACKUP_NAME="hiresense_backup_${TIMESTAMP}.sql.gz.enc"

# Execute compressed PostgreSQL dump
pg_dump -h db.supabase.co -U postgres -d postgres | gzip > /tmp/${BACKUP_NAME}

# Encrypt backup archive using openssl AES-256
openssl enc -aes-256-cbc -salt -pbkdf2 -in /tmp/${BACKUP_NAME} -out /tmp/${ENC_BACKUP_NAME}
-pass file:/etc/backup_pass.key

# Push to Cloudflare R2 using AWS CLI profile
aws s3 cp /tmp/${ENC_BACKUP_NAME} s3://hiresense-backups-bucket/${ENC_BACKUP_NAME}
--endpoint-url https://<account_id>.r2.cloudflarestorage.com

# Cleanup temporary files
rm /tmp/${BACKUP_NAME} /tmp/${ENC_BACKUP_NAME}
```

This ensures full disaster recovery under strict standard operating environments.